

Scalability & Future Plan

Date	04 July 2026
Team ID	XXXXXX
Project Name	XXXXXX
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Dependence on Static Historical Training Data	The model may fail to capture sudden real-time economic shifts, changes in inflation rates, or updated banking policies without manual retraining.	High
2	Lack of Dynamic Real-Time Bureau Integration	The system currently relies on manual user inputs for credit scores and financial history instead of live API fetching from an official credit bureau	Medium
3	Black-Box Decision Predictions	The system provides loan approval/rejection predictions but lacks native Explainable AI features (like SHAP or LIME) to tell the user exactly why their application was rejected	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Handles a single user or small concurrent team local requests	Deploy on a cloud infrastructure (like AWS or Azure) using Docker containers managed by Kubernetes with automated horizontal autoscaling to handle thousands of simultaneous loan applicants.
2	Data Storage	Uses file-based storage (like CSVs) or a lightweight local relational database (SQLite) for storing loan profiles and applicant information	Migrate to an enterprise cloud database tier (like PostgreSQL or Amazon RDS) configured with read-replicas, coupled with a NoSQL database (like MongoDB) to comfortably process unstructured applicant logs.
3	Performance	Machine Learning model predictions run synchronously on the main thread, which can bottleneck when processing large batches of applications.	Implement asynchronous processing by decoupling the ML inference engine using a task queue framework (like Celery with Redis). Implement model caching for repeating credit profiles to provide instantaneous results.
4	Security	Employs basic development-level authentication and handles sensitive applicant financial data in plaintext or	Implement banking-grade security protocols including OAuth 2.0 with JWT authentication, end-to-end data encryption (AES-256 for data at rest and TLS 1.3 for data in transit), and strict compliance monitoring."

		basic local hashing.	
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Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Live Credit Bureau API Integration & Explainable AIConnect the core system directly to live financial credit scoring APIs (e.g., CIBIL, Experian) to automate data verification, and integrate SHAP/LIME frameworks.	Next 3 Months	Minimizes user manual data entry errors, entirely eliminates fraudulent score claims, and provides transparent, regulation-compliant explanations for why a loan was approved or denied.
Phase 3	Alternative Data Scoring & Multi-Model EnsemblingIncorporate non-traditional predictive data points (utility bill trends, digital transaction habits) and implement real-time model retraining pipelines.	Next 4 to 6 Months	Expands the target market by accurately scoring ""thin-file"" borrowers (applicants without deep traditional banking history) while keeping model accuracy continuously optimized against market shifts.
Phase 4	Microservices Migration & Mobile Application LaunchBreak down the web monolith into resilient microservices and deploy native iOS/Android cross-platform apps using Flutter/React Native.	Next 7 to 12 Months	Ensures maximum architectural fault tolerance, significantly boosts user accessibility for on-the-go loan applications, and scales seamlessly to support high-volume enterprise banking operations.